Diet Contributes Significantly to the Body Burden of PBDEs in the General U.S. Population

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Background Exposure of the U.S. population to polybrominated diphenyl ethers (PBDEs) is thought to be via exposure to dust and diet. However, little work has been done to empirically link body burdens of these compounds to either route of exposure. Objectives The primary goal of this research was to evaluate the dietary contribution to PBDE body burdens in the United States by linking serum levels to food intake. Methods We used two dietary instruments—a 24-hr food recall (24FR) and a 1-year food frequency questionnaire (FFQ)—to examine food intake among participants of the 2003–2004 National Health and Nutrition Examination Survey. We regressed serum concentrations of five PBDEs (BDE congeners 28, 47, 99, 100, and 153) and their sum (∑PBDE) against diet variables while adjusting for age, sex, race/ethnicity, income, and body mass index. Results ∑PBDE serum concentrations among vegetarians were 23% ($p = 0.006$) and 27% ($p = 0.009$) lower than among omnivores for 24FR and 1-year FFQ, respectively. Serum levels of five PBDE congeners were associated with consumption of poultry fat: Low, medium, and high intake corresponded to geometric mean ∑PBDE concentrations of 40.6, 41.9, and 48.3 ng/g lipid, respectively ($p = 0.0005$). We observed similar trends for red meat fat, which were statistically significant for BDE-100 and BDE-153. No association was observed between serum PBDEs and consumption of dairy or fish. Results were similar for both dietary instruments but were more robust using 24FR. Conclusions Intake of contaminated poultry and red meat contributes significantly to PBDE body burdens in the United States.